

Integrated project report — Hong Kong thermal associations with first hospitalisation after CHD or HF diagnosis, 2013–2023

Bob Shen Ruililin

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- **Audience:** Professor David Bishai (supervisor), with shared scientific use by Roro, Hogan, and Bob.
 - **Date:** 10 August 2026.
 - **Mode:** Supervisor-facing scientific report (internal). Gate language and collaborator roles are permitted here.
 - **Provenance:** Health estimates are HA_APPROVEDAggregate; calibration is SYNTHETIC_CALIBRATION; weather reproduction checks are REAL_PUBLIC_HK0.
 - **Companion packet:** journal manuscript, supplement, release tables and figures, and the redacted scientific contract; paths are indexed in Appendix A.
 - **Scientific refusals:** no invented human decisions, agent Gate 3 freeze, committed HA microdata, or synthetic coefficients presented as findings.
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1. One-page executive summary

This project asked whether monthly thermal burden in Hong Kong is associated with governed monthly counts of first cardiovascular hospitalisation among people with type 2 diabetes and/or hypertension. The delivered outcomes are first hospitalisations after a first coronary heart disease (CHD) or heart failure (HF) diagnosis record during January 2013–December 2023 (132 territory-months). Admission cause is not observed. Stroke was named in covering correspondence but was not attached. The scientifically honest estimand is therefore an ecological monthly count ratio under a cohort-restricted first-event construction, not principal-diagnosis incidence and not a daily triggering analysis.

The analysis-of-record architecture fits six exposures separately for each outcome: mean temperature, mean maximum temperature, mean minimum temperature, official hot nights, official cold days, and official very hot days. Models are negative binomial with calendar-month factors, a natural spline of time (4 df), and a days-in-month offset. Continuity reporting uses Newey–West lag-6 intervals together with Model, HC1, and Newey–West lag-3 intervals. No interval is selected because it excludes the null.

The twelve continuity estimates are complete and internally validated. The two smallest unadjusted p-values are CHD hot nights (count ratio 1.022; NW6 1.002–1.042; $q = 0.192$) and HF cold days (1.073; NW6 1.006–1.144; $q = 0.192$). For CHD hot nights, Model and HC1 intervals include the null (0.995–1.049 and 0.997–1.047). All twelve Benjamini–Hochberg q-values exceed 0.19. CHD residual autocorrelation at lag 1 is about 0.51; HF is about 0.15. Joint entry of correlated heat metrics inflates the CHD hot-night coefficient relative to the separate model. These facts support exploratory signals inside a definition- and uncertainty-aware panel. They do not support a Gate 3 confirmatory headline of “CHD-hot / HF-cold discovery.”

A constrained monthly-outcome / daily-exposure (M|D) method was calibrated on synthetic outcomes. The F1.0 pilot failed because HF-like null cells absorbed residual cold-season structure.

F1.1 corrected nuisance seasonality. F1.2 recomputed gates on worst cells rather than averages. The 500-replicate decision is fail: null Type I 0.048–0.150, minimum coverage 0.840, maximum non-null relative bias 32.77, maximum moderate false-sign 0.808, stress coverage 0.842. No real daily coefficient is admitted.

What is learned: the data contract; source-locked weather morphology; a reproducible monthly panel with uncertainty discordance; and a documented refusal of uncalibrated daily recovery. What is not learned: stroke associations; cohort incidence; principal-diagnosis effects; protected primary thermal claims; or daily lag structure from monthly sums. The lead scientific recommendation is Option A (explicit no confirmatory primary; methods-focused exploratory paper); Gate 3 approval remains human-owned. Ranked decisions for humans follow in Section 10.

2. First-principles question

The first-principles question is not “which p-value is smallest.” It is:

Given governed monthly first-hospitalisation counts without admission cause, and daily weather that can be aggregated into several non-equivalent monthly thermal burdens, what associations can be estimated honestly, and which stronger claims must be refused?

That question forces five distinctions:

1. **Outcome construction** versus disease label in ordinary language. A first hospitalisation after a first CHD diagnosis record is not an admission for CHD in the principal-diagnosis sense.
2. **Monthly count ratio** versus cohort incidence-rate ratio. The analysis uses a days-in-month offset. A general-population offset remains an ecological sensitivity because the T2D/HTN risk set still eligible for a first event is unavailable.
3. **Thermal definition** versus a single temperature number. Mean temperature, hot nights, cold days, spells, and heat-month tails select different months (Wang et al. 2019; Li et al. 2025; Y. Guo et al. 2017).
4. **Daily mortality attributable fraction / modelled excess deaths** versus monthly morbidity associations (J. Liu et al. 2020; Z. Liu et al. 2026).
5. **Methods feasibility** versus health finding. An M|D estimator that fails calibration does not become a daily result (Basagaña and Ballester 2024).

The project’s scientific product is therefore a contract-respecting panel with explicit refusals, not a press-ready binary story.

A useful corollary follows for supervision. When a monthly ecological design is forced by governance, the correct response is not to invent daily resolution in the prose. The correct response is to state the monthly estimand, stress-test thermal definitions, report residual dependence and standard-error discordance, and treat any attempt to recover daily coefficients as a gated methods question. That posture is what this report means by aggregation-aware inference.

2.1 Scientific boundaries that do not move without new data

Several boundaries are already closed by the delivered files and do not await Gate 3:

1. **No AMI / principal-diagnosis claims** from this extract. Admission cause is absent.
2. **No stroke results** until a stroke file arrives. Naming stroke in correspondence is not delivery.
3. **No cohort incidence language** without the still-at-risk denominator.
4. **No rich daily DLNM** at $n = 132$ months after the constrained M|D route failed calibration.

5. **No selection of SE method by null exclusion.** Continuity NW6 is displayed with the full ladder.
6. **No promotion of synthetic calibration** into health findings.

These boundaries are scientific, not administrative. Crossing them would create false findings even if the writing were elegant.

3. Concise chronology from correspondence (internal evidence)

This chronology summarises the scientific path from the nine-page Outlook thread (21 April-6 August 2026) and the 7 August data receipt. It is internal provenance, not a References list. Email addresses and private PDF quotations are omitted.

21 April-9 July 2026. Correspondence concerned programme logistics and the absence of a research update; no health outcome was delivered in that period.

10-12 July 2026. Bob reported a validated Hong Kong Observatory pipeline and stated that synthetic health outcomes had tested code only. He asked for diagnosis coding, principal-diagnosis availability, episode rules, age bands, suppression, and output vetting. Professor Bishai proposed merging monthly temperature with monthly AMI and stroke admission risk and requested a descriptive Table 1 plus patient-level adjustment ambition (age, sex, medications, BMI). Roro confirmed ICD-9 coding, DAE inclusion, complete admission outcomes at patient level in that extract, no ED-to-inpatient episodes in that extract, age 65-69 / 70-74 distinguishability, and controlled HPC access. Roro asked whether additional HA work required an IRB update; no PI answer appears in the supplied thread.

17-22 July 2026. The project moved away from principal-diagnosis AMI claims when the general HA file was found not to contain admission causes. Near-term focus shifted toward monthly stroke aggregates. Daily weather morphology inside monthly exposures was retained; an ordinary monthly lag was not to be described as a daily DLNM. Hogan proposed source-locking a heatwave definition, counting events by month, and defining hot months by the upper tail of that count distribution.

22-28 July 2026. Roles clarified: Hogan on weather methods and writing mentorship; Roro on regression oversight and outcome construction; Bob on exposure pipeline, integration, and remaining Methods. Hogan shared a live manuscript and wrote the weather Methods. That live file remains collaborative manuscript authority for weather text.

6-7 August 2026. Roro delivered governed monthly CHD and HF aggregates for a T2D/HTN 2013-2023 cohort. Because admission cause was absent, the event was defined as first hospitalisation after the patient's first diagnosis record for the cardiovascular condition, with assistance from Dr Jingjing Zhou on first-diagnosis records. Stroke was named but not attached. Gate 1 closed conditionally for CHD/HF; Gate 2 closed for CHD/HF territory-month first-event aggregates; Gate 3 stayed open.

10 August 2026. Final exploratory protocol, claim tiers, uncertainty ladder, weather source validation, and M|D calibration F1.0→F1.1→F1.2 were frozen or completed as post-outcome methods work. Release validation passed 29/29 checks. No agent Gate 3 freeze occurred.

The path is planned AMI/stroke → no admission cause → delivered CHD/HF first-event pivot. The pivot is a data-contract change, not a scientific failure of collaborators.

4. Exact data / outcome / denominator / estimand contract

Element	Contract
Period	January 2013–December 2023; 132 territory-months
Cohort universe	People diagnosed with T2D and/or HTN during 2013–2023
CHD event	First hospitalisation after first CHD diagnosis record
HF event	First hospitalisation after first HF diagnosis record
Admission cause	Not recorded
Stroke	Not delivered
Age/sex strata	Not delivered (capability $\neq$ delivery)
Grain	Territory $\times$ calendar month
Provenance label	HA_APPROVED_AGGREGATE
Analysis-of-record offset	$\log(\text{days in month})$
Population $\times$ days offset	Ecological sensitivity only
Target parameter	$\exp(\beta)$: monthly count ratio
Not the target	Cohort IRR; principal-diagnosis RR; AMI; daily DLNM RR; mortality AF

Totals (outputs/release_chd_hf/tables/table1_outcome_summary.csv): CHD 156,156 events (mean 1,183.0/month); HF 29,681 (mean 224.9/month).

Governance: source xlsx and merged panels are gitignored. Aggregation reduces re-identification risk but does not grant external-publication authority. Written Roro/Bishai confirmation remains required before submission or public promotion (reports/data_receipt_2026-08-07.md).

5. Source-locked weather morphology and five reproduction checks

Daily HKO Headquarters observations create monthly means and official extreme-day counts. Official thresholds used in the continuity panel are hot night ($T_{\min} \geq 28^\circ\text{C}$), very hot day ($T_{\max} \geq 33^\circ\text{C}$), and cold day ($T_{\min} \leq 12^\circ\text{C}$). Spell and atmospheric morphologies follow Wang et al. (2019) and Li et al. (2025) after source lock (Wang et al. 2019; Li et al. 2025).

Five reproduction checks all passed (outputs/release_chd_hf/supplement/methods_feasibility/weather_det

Check	Expected	Observed
Li events 1980–2023	57	57
Wang VHD days 2006–2015	204	204
Wang HN days 2006–2015	230	230
Wang VHD ≥ 5 event-days 2006–2015	48	48
Wang HN ≥ 5 event-days 2006–2015	56	56

Calendar spillover morphology (event starts, overlapping events, event-days, same-day and post-event phases) is an exposure audit, not twelve independent discovery tests. Provisional hot-month / cold-month indicators remain unlocked pending Hogan’s reference-period decision; their estimates stay supplementary.

Key real-data figures

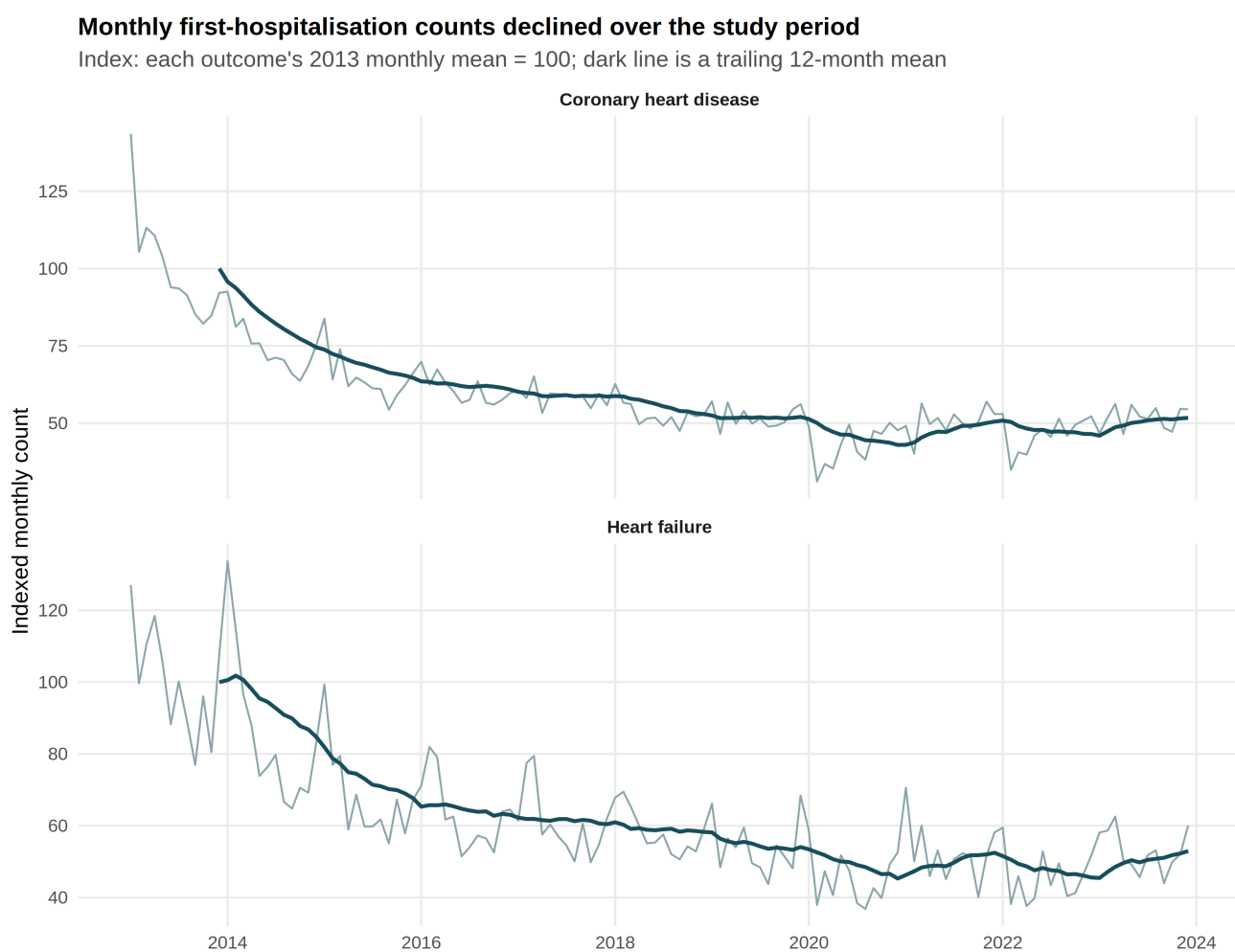


Figure 1: Figure 1. Indexed monthly first-hospitalisation series for CHD and HF (HA_APPROVED_AGGREGATE).

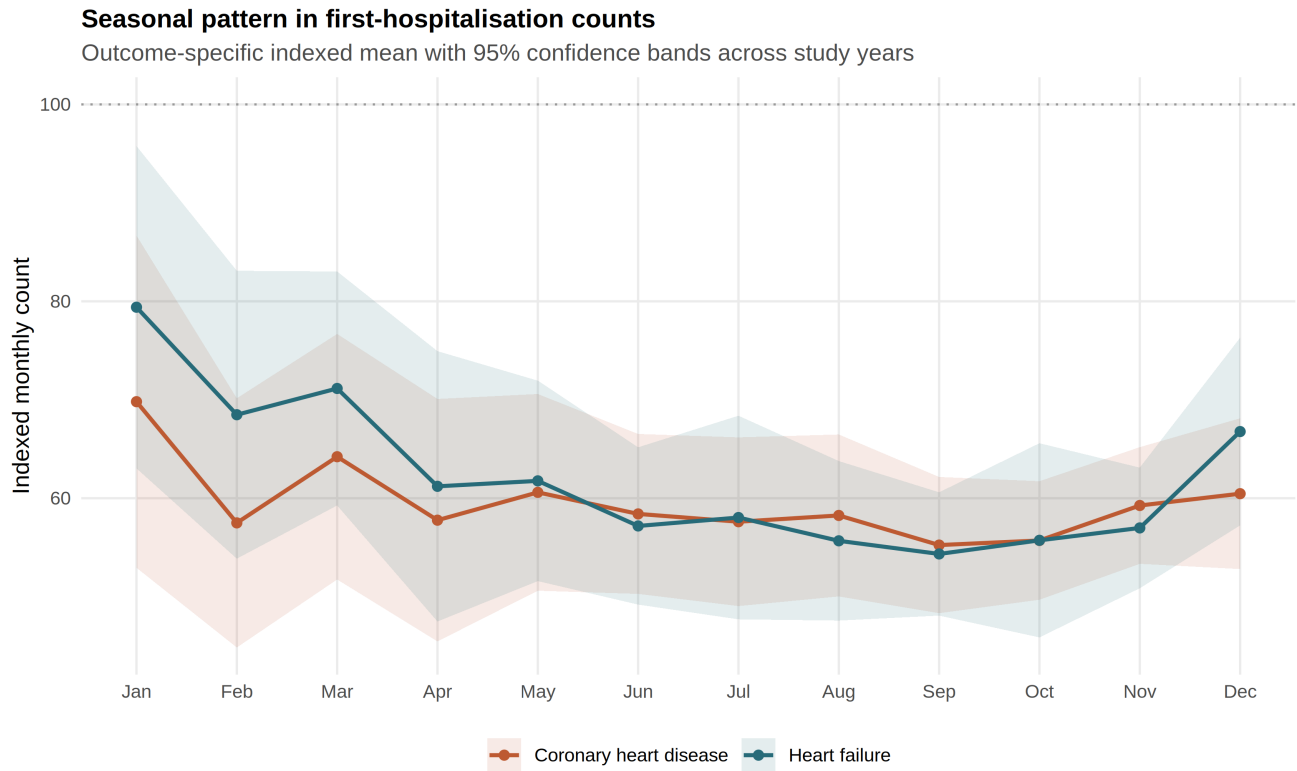


Figure 2: Figure 2. Seasonal profiles of indexed CHD and HF first-hospitalisation counts.

6. Real CHD/HF analysis architecture and results

6.1 Architecture

Amended core: separate single-exposure negative-binomial models for CHD and HF $\times$ six exposures (twelve contrasts). Controls: calendar-month factor + ns(time, 4). Offset: days. Continuity SE: Newey-West lag 6, displayed with Model / HC1 / NW3 (outputs/release_chd_hf/tables/table4_uncertainty). Figure 5). Multiplicity: Benjamini-Hochberg across the twelve NW6 p-values. Claim ledger: CVD-01...CVD-12 (claim_ledger_v2.csv). Release validation: 29/29 pass (outputs/reports/cvd_real_release_val).

6.2 Complete nulls and core estimates

All twelve continuity estimates (table2_core_models.csv):

ID	Outcome	Contrast	RR (NW6 95% CI)	p	q
CVD-01	CHD	Mean temp /1 °C	0.993 (0.976- 1.010)	0.424	0.726
CVD-02	CHD	Mean Tmax /1 °C	0.993 (0.979- 1.007)	0.321	0.642
CVD-03	CHD	Mean Tmin /1 °C	0.994 (0.977- 1.012)	0.509	0.764

ID	Outcome	Contrast	RR (NW6 95% CI)	p	q
CVD-04	CHD	Hot nights /5	1.022 (1.002- 1.042)	0.032	0.192
CVD-05	CHD	Cold days /5	0.995 (0.949- 1.043)	0.823	0.900
CVD-06	CHD	Very hot days /5	0.999 (0.974- 1.025)	0.962	0.962
CVD-07	HF	Mean temp /1 °C	0.974 (0.947- 1.002)	0.069	0.207
CVD-08	HF	Mean Tmax /1 °C	0.981 (0.958- 1.005)	0.113	0.272
CVD-09	HF	Mean Tmin /1 °C	0.973 (0.947- 1.000)	0.050	0.202
CVD-10	HF	Hot nights /5	1.003 (0.976- 1.031)	0.825	0.900
CVD-11	HF	Cold days /5	1.073 (1.006- 1.144)	0.031	0.192
CVD-12	HF	Very hot days /5	0.995 (0.963- 1.028)	0.764	0.900

Complete nulls / near-nulls under continuity NW6: CHD mean temperatures and cold/very-hot days; HF hot nights and very hot days; HF mean temperatures near null with intervals that touch or include 1 under NW6. **Exploratory unadjusted signals:** CVD-04 and CVD-11. **Multi-plicity:** all $q > 0.19$.

Figures: indexed series (Figure 1), seasonality (Figure 2), core forest (Figure 3) under out-puts/release_chd_hf/figures/.

6.3 Standard-error ladder (explicit key contrasts)

CHD hot nights, $RR = 1.022$:

SE method	95% CI
Model	0.995-1.049
HC1	0.997-1.047
NW3	1.0003-1.0439
NW6	1.002-1.042

HF cold days, $RR = 1.073$:

SE method	95% CI
Model	1.023-1.125

SE method	95% CI
HC1	1.011-1.138
NW3	1.007-1.143
NW6	1.006-1.144

NW6 is continuity analysis of record with the full ladder shown. It is not designated primary because an interval excludes the null ([Lazarus et al. 2018](#)). For CHD hot nights, Model and HC1 include the null; NW3 and NW6 exclude it on the unrounded scale. That discordance is a result.

6.4 Multiplicity and residual dependence

Benjamini-Hochberg q-values across the twelve amended-core NW6 contrasts are all greater than 0.19. The two unadjusted associations below conventional 0.05 therefore remain exploratory panel results.

Pearson residual ACF1 ≈ 0.51 for CHD core models (range 0.508-0.534) and ≈ 0.15 for HF (range 0.131-0.179). Ljung-Box tests reject white noise for CHD, not for HF. Negative-binomial IN-GARCH(1,1) fits without an offset reduce residual autocorrelation and retain positive CHD hot-night and HF cold-day associations; these unoffset diagnostics are not directly comparable with days-offset continuity estimates. Serial dependence is part of the inferential problem.

6.5 Joint models

After month/trend residualisation, VIF = 4.66 for joint mean Tmax/Tmin and ≈ 1.96 for hot nights / very hot days. Joint extreme-day modelling raises the CHD hot-night ratio from 1.022 to 1.045 (1.015-1.075). HF cold days remain about 1.073. Joint models are collinearity diagnostics, not preferred estimates.

6.6 Trend, depletion, lag, influence

Across trend/depletion scenarios (table3_robustness_summary.csv; Figure 4), CHD hot nights range ≈ 1.011 -1.025; HF cold days ≈ 1.043 -1.113. Pre-2020 HF cold days: 1.113 (1.053-1.176). Lag-1/2 attenuate CHD hot nights toward the null; HF cold remains elevated at lag 1 (1.073) and weaker at lag 2 (1.053). Influence exclusions (CHD 2020-02; HF 2022-02 for cold days) leave directions unchanged.

6.7 Specification discipline and what was not searched

The final exploratory protocol freezes a finite model universe after outcomes were visible (analysis_plan/final_reanalysis_protocol_2026-08-10.md). That status matters. The panel is post-outcome exploratory, not a confirmatory preregistration. Several rules follow:

- The twelve amended-core contrasts form one multiplicity family.
- Spillover morphology variants are an exposure audit family; they cannot rescue a core claim.
- Warm-season heat and cold-season cold cut-points are not searched after fitting.
- A formal cross-outcome hot-versus-cold interaction was not estimated in the completed real-data release; separate coefficients do not establish $\Delta \neq 0$.

SAP Amendment A1 already corrected an earlier tendency to treat joint P02/P04 as headline candidates and to interpret population $\times$ days as a true risk-set offset (reports/gate3_decision_packet_2026-08-07.md). The present report inherits those corrections.

6.8 Reading the exploratory CHD hot-night / HF cold-day pattern without over-claiming

A qualitative panel pattern is visible: CHD first-hospitalisation counts track official hot-night burden more closely than mean temperature or cold days; HF counts track cold-day burden more

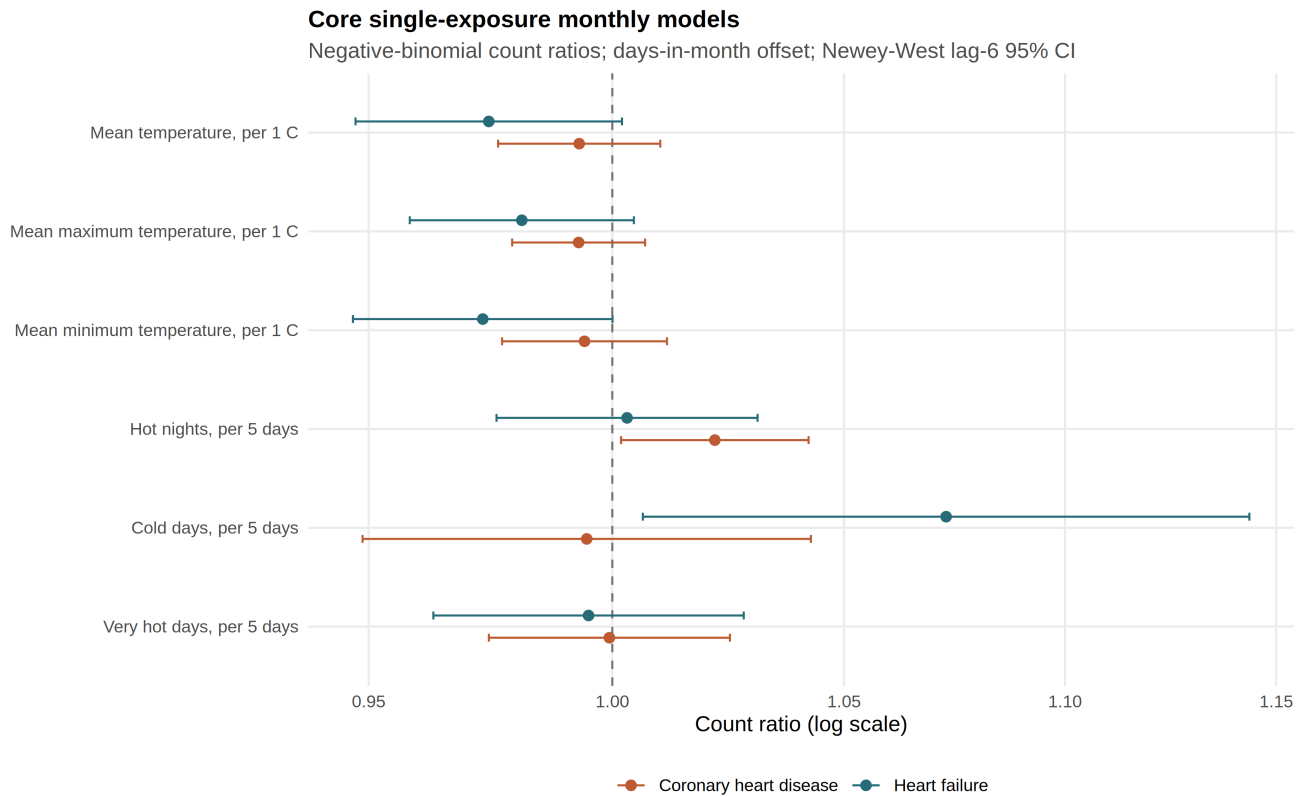


Figure 3: Figure 3. Continuity-panel forest plot (days offset; Newey-West lag 6). All twelve q-values > 0.19.

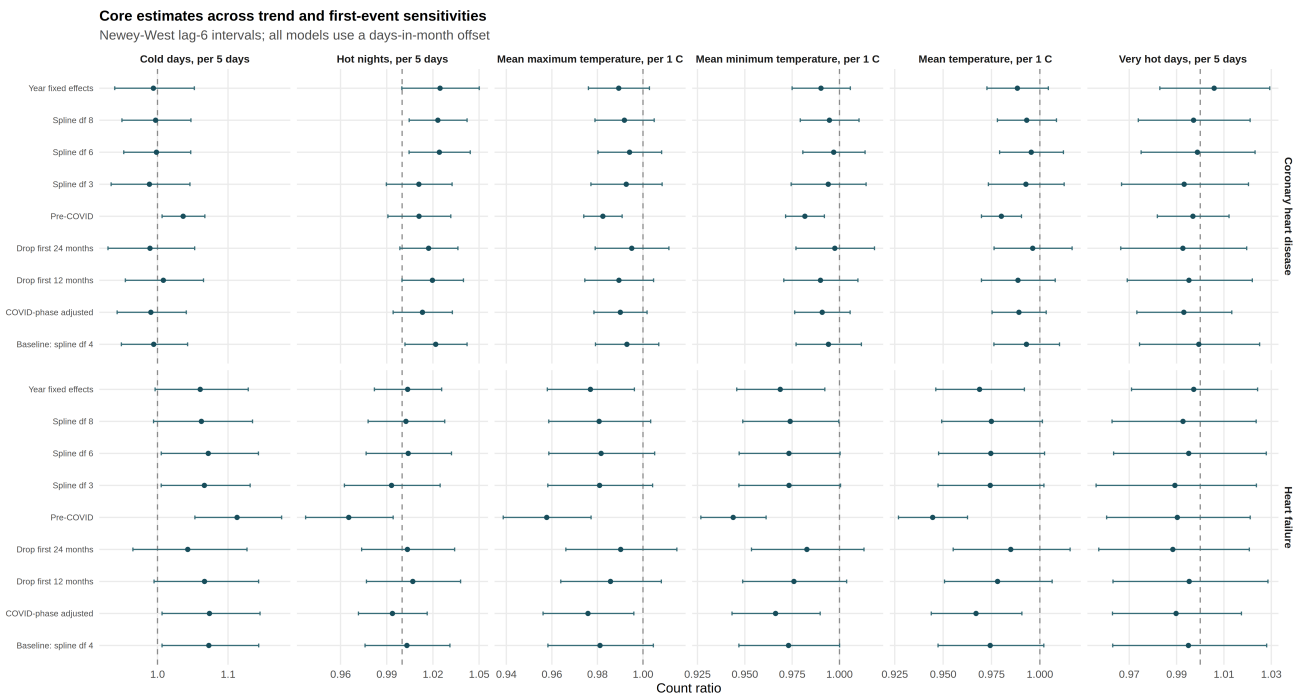


Figure 4: Figure 4. Trend and first-event-depletion sensitivity forest.

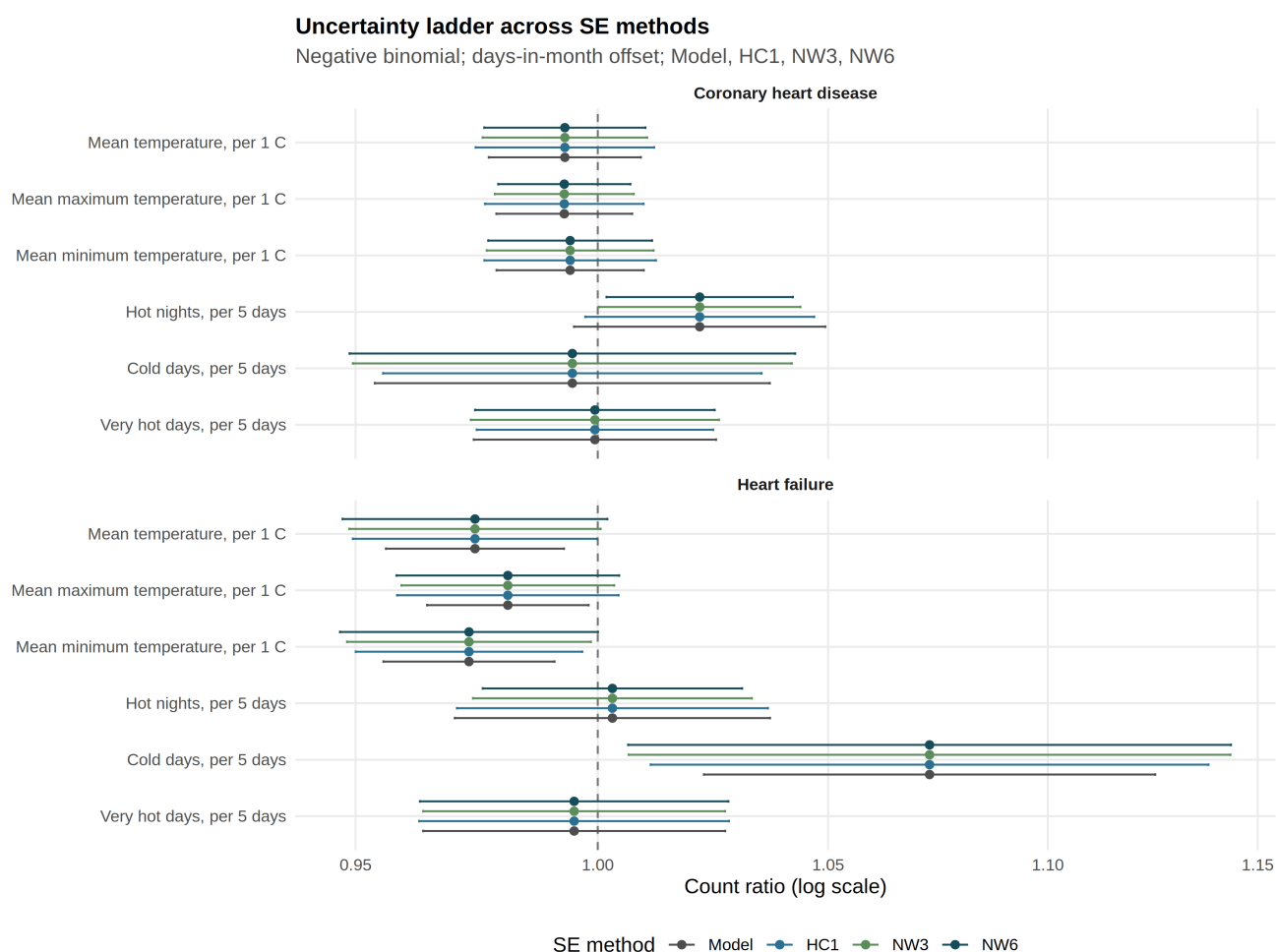


Figure 5: Figure 5. Standard-error method ladder (Model / HC1 / NW3 / NW6) for the twelve continuity contrasts.

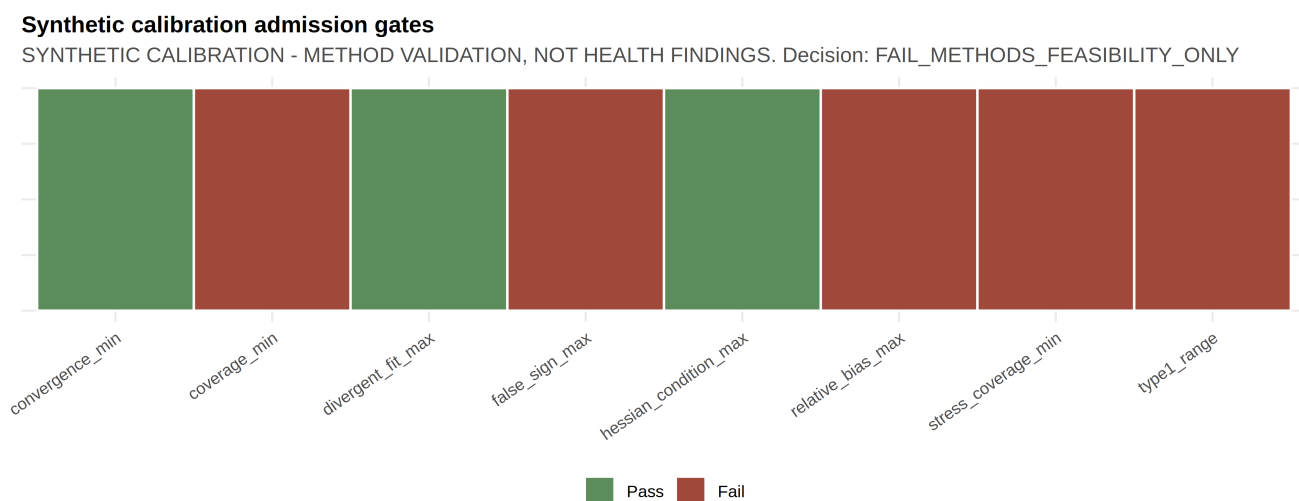


Figure 6: Methods validation only — F1.2 synthetic M|D gate pass/fail summary. Not a CHD/HF health finding.

closely than hot nights. It is consistent with historical Hong Kong cold evidence and contemporary concern about nighttime heat (Goggins, Chan, and Yang 2013; Y. T. Guo et al. 2024), but it is not confirmatory. The complete panel should retain CVD-04 and CVD-11 while also retaining $q = 0.192$, CHD Model/HC1 intervals that include the null, joint-model inflation, residual dependence, and the failed M|D calibration. That is the boundary between hiding a signal and manufacturing a discovery.

7. M|D rationale, F1.0 failure, F1.1 correction, F1.2 worst-cell failure

Monthly health sums erase day-of-month timing. Basagaña and Ballester show that aggregated-outcome likelihoods can, under conditions, recover daily temperature-mortality relations from temporally aggregated outcomes, with monthly recovery more variance-prone than weekly (Basagaña and Ballester 2024, 2026). The project therefore tested a constrained M|D estimator as methods feasibility, not as a default DLNM.

F1.0 (100-rep pilot). Daily nuisance seasonality used one sine/cosine pair. CHD-like nulls looked acceptable. HF-like null Type I rose to 0.62–0.75 with coverage 0.25–0.38. Residual cold-season structure was absorbed by the cold exposure. This was a design failure, not a real HF finding (md_pilot_f1_0_failure_note.md).

F1.1. Nuisance seasonality was amended to calendar-month indicators, matching the analysis of record, inside the frozen 20-parameter cap. The failed pilot was retained for audit. No real coefficient was inspected to choose the amendment.

F1.2 (500-rep worst-cell gates). Averaging across cells can hide a failed HF-like cell behind a calibrated CHD-like cell. Gates were recomputed on binding worst cells without relaxing thresholds or fitting new real models. Decision: FAIL_METHODS_FEASIBILITY_ONLY. Null Type I 0.048–0.150; min coverage 0.840; max non-null relative bias 32.77; max moderate false-sign 0.808; stress coverage 0.842. Convergence/Hessian/divergence passed; Type I, coverage, bias, false-sign, and stress coverage did not.

Admission rule. Because gates failed, no real daily-exposure coefficient enters Results. The monthly panel remains analysis of record.

7.1 Why the failure is scientifically useful

A failed calibration is not wasted work. It answers a concrete methods question that supervisors and reviewers will ask: given only monthly sums, can a constrained daily-exposure model be trusted here? Under the frozen gates and this synthetic design, the answer is no. That refusal protects the manuscript from a common failure mode in environmental epidemiology—borrowing daily language for monthly data.

The failure also clarifies next designs. If daily or weekly governed outcomes become available, a conventional time-series model is the honest route. If only monthly sums remain available, the analysis should stay monthly and invest in definition robustness, residual dependence, and denominator improvement rather than in uncalibrated daily recovery.

7.2 Comparator noncomparability (F1.2)

F1.2 also records that comparator models whose coefficients target a different monthly quantity are not scored against the daily M|D β . Their convergence and predictive metrics remain reportable, but they cannot be used to invent a pass for the daily method. This prevents a second subtle misuse: treating a well-behaved monthly negative-binomial comparator as evidence that the daily coefficient was recovered.

8. Jasmine daily mortality AF → Roro modelled excess deaths → current monthly morbidity map

Three claim classes must stay distinct (literature/final_methods_evidence_map.md):

1. **Jingwen Liu et al. (2020)** — daily Hong Kong mortality, 2006–2016; DLNM + quasi-Poisson; cold AF 4.72% vs heat 0.16%; moderate AF 4.25% vs extreme 0.63% ([J. Liu et al. 2020](#)). These are mortality attributable fractions.
2. **Zhenyuan Liu et al. (2026, medRxiv)** — model-based excess deaths 2014–2023 under four heatwave scenarios using transported local RRs; range 1,455–3,238 across definitions ([Z. Liu et al. 2026](#)). These are modelled excess deaths, not new daily regressions and not hospitalisation coefficients. Public cite = medRxiv v1; private revised PDF is internal audit only.
3. **This project** — monthly CHD/HF first-hospitalisation count ratios under T2D/HTN first-event construction. Continuity exploratory signals exist; multiplicity-protected primary claims do not.

The family is cumulative in literature motivation: daily mortality AF → multi-definition modelled heat deaths → monthly morbidity associations. It is not a licence to paste AF or excess-death numbers into Table 2.

8.1 How each class should appear in team prose

Class	Where it belongs	Sentence template
Daily mortality AF	Introduction / evidence spine	“Jingwen Liu et al. (2020) reported cold AF 4.72% versus heat 0.16% for Hong Kong mortality.”
Modelled excess deaths	Companion mortality baseline	“Zhenyuan Liu et al. (2026) estimated 1,455–3,238 model-based excess deaths across four heatwave definitions.”
Monthly morbidity count ratios	Results tables	“Under days-in-month offset, the continuity count ratio for HF cold days was 1.073 (NW6 1.006–1.144; $q = 0.192$).”
M D validation	Methods / supplement	“Synthetic calibration failed F1.2 worst-cell gates; no real daily coefficient was admitted.”

If a draft sentence mixes two rows of this table, rewrite it.

8.2 Local morbidity precedents that still do not rewrite the estimand

Goggins et al. (2012, 2013) and Chan et al. (2013) provide historical daily Hong Kong morbidity baselines for stroke, AMI, and broad admissions ([Goggins et al. 2012](#); [Goggins, Chan, and Yang 2013](#); [Chan et al. 2013](#)). Yang et al. (2025) support a pre-specified influenza sensitivity for cold models in longer weekly stroke series ([Yang et al. 2025](#)). Guo et al. (2024, 2025) motivate hot-night intensity and staged pollution adjustment ([Y. T. Guo et al. 2024, 2025](#)). None of these studies converts the present CHD/HF first-event monthly aggregates into principal-diagnosis or daily-trigger findings. They motivate symmetry, covariates, and caution.

9. What is scientifically learned versus not learned

Learned (established / supported)

- Exact outcome and denominator contract for the delivered files.
- Source-locked weather morphology with five passing reproduction checks.
- A complete twelve-contrast monthly panel with claim-ledger identity and release validation.
- Uncertainty discordance: CHD hot-night intervals cross the null under Model/HC1 but not under continuity NW6.
- Multiplicity: all core $q > 0.19$.
- Residual dependence: CHD ACF1 ≈ 0.51 ; HF ≈ 0.15 .
- Joint-model inflation of CHD hot nights as a collinearity warning.
- Robustness ranges for trend, depletion, lag, and influence.
- Documented MJD methods failure under F1.2 worst-cell gates.

Not learned (future / blocked / hypothesis-only)

- Stroke associations.
- Principal-diagnosis CHD/HF or AMI effects.
- True cohort incidence with the still-at-risk denominator.
- Age, sex, medication, or subtype modification.
- Locked hot-month / cold-month primary effects.
- Daily lag-specific triggering from monthly sums.
- A Gate 3 confirmatory headline.
- Causal pathway stories (nocturnal recovery impairment; cold-triggered decompensation) beyond hypothesis language.
- External publication permission.

9.1 Claim-tier map for this report

Using `analysis_plan/final_claim_tiers.yml`:

Tier	Examples in this report	Manuscript handling
Established	132 months; first-event construction; no admission cause; stroke absent; Gate 3 open; weather checks pass	Allowed
Supported	All twelve continuity estimates; VIF; residual ACF; robustness ranges; SE ladder	Allowed with uncertainty and model IDs
Exploratory	CHD hot-night and HF cold-day signals; joint-model inflation	Allowed with exploratory qualifier and multiplicity
Hypothesis	Nocturnal recovery impairment; cold-triggered HF decompensation; depletion as care-pattern change	Discussion only
Future	Stroke; cohort incidence; age/sex/medication; principal-diagnosis; daily DLNM	Limitations / next-data only

Forbidden promotions remain binding: calibration $\rightarrow$ real finding; exploratory $\rightarrow$ primary without Gate 3; mortality AF $\rightarrow$ monthly morbidity; excess deaths $\rightarrow$ hospitalisation burden; population offset $\rightarrow$ cohort incidence.

10. Ranked decisions for humans

Decision face for the supervision meeting

Decision	Lead recommendation	Human owner / response
Gate 3 posture	Option A: no confirmatory primary; methods-focused exploratory paper	Team: A / B / C
External dissemination	Hold until written aggregate-release approval	Bishai + Roro: approve / hold
IRB/protocol amendment	Record a PI determination before submission	Bishai: required / not required / seek advice
Stroke	Deliver under a documented first-event contract or declare out of this paper	Roro + team
Cohort denominator	Request monthly T2D/HTN members still eligible for a first event	Roro + Bishai
Weather HM/CM	Keep supplementary until the reference rule is locked	Hogan + team
Authorship	Resolve order, corresponding author, co-first wording, and Zhou role in writing	All proposed authors

For Professor Bishai

1. Decide Gate 3 posture: freeze no confirmatory primary; or freeze a carefully worded exploratory panel summary; or request additional governed data before any headline.
2. Decide whether IRB/protocol amendment is required for the HA work (question raised; unanswered in supplied thread).
3. Confirm external dissemination permission for disclosure-minimised aggregates.
4. Set authorship expectations with Roro and Hogan (order; corresponding author; co-first suggestion).

For Roro

1. Deliver stroke file or confirm permanent absence for this manuscript cycle.
2. Confirm ICD lists and *_inpatient semantics.
3. Advise whether the T2D/HTN cohort still at risk of a first event can be obtained.
4. Confirm health-data Methods text and publication authority for aggregates.
5. Resolve co-first / author-order discussion in writing.

For Hogan

1. Lock weather reference period and HM/CM starters (or explicitly keep them supplementary).
2. Review whether live weather Methods enter the journal manuscript under his text.
3. Advise on writing register for the journal draft (already applied here in sparse academic English).

For Bob

1. Keep NW6 as continuity analysis of record with full SE ladder; do not p-hunt intervals.

2. Do not promote CVD-04/CVD-11 to primary without Gate 3.
3. Keep M|D in methods-feasibility quarantine.
4. Sync repository prose with Hogan’s live manuscript without overwriting weather Methods.
5. Maintain provenance firewall and claim-ledger comments in manuscript drafts.

10.1 Lead scientific recommendation and Gate 3 menu

Lead recommendation (scientific, not a freeze): Option A — explicit no confirmatory primary; methods-focused exploratory paper. Report CVD-01 to CVD-12 with q-values and the full SE ladder; refuse a single headline differential thermal claim. This matches current multiplicity (all $q > 0.19$), CHD hot-night SE sensitivity, and the failed M|D calibration. Team approval of Gate 3 remains human-owned.

Alternatives:

Option B — Exploratory panel summary without discovery framing. Allow a qualitative sentence that CHD tracks hot nights more than means/cold and HF tracks cold days more than hot nights, while stating all $q > 0.19$ and Model/HC1 discordance for CHD hot nights. Still not a confirmatory primary.

Option C — Defer headline until next data. Hold external manuscript submission until stroke and/or risk-set denominators arrive. Internal report and methods quarantine remain usable now. The freeze itself is not executed here.

11. Technically credible next-data programme

A credible next programme is incremental and governance-first:

1. **Stroke delivery:** territory-month counts for January 2013–December 2023 under the same first-event contract, with SHA-256, schema, suppression rule, and event-construction text; otherwise an explicit decision that stroke is outside this paper.
2. **Dictionary lock:** exact ICD lists, the meaning of *_inpatient, transfer/readmission handling, suppression rules, and any ED inclusion.
3. **Risk-set denominator:** one row per month with the number of T2D/HTN cohort members still eligible for a first CHD or HF hospitalisation, including cohort entry, censoring, and suppression rules.
4. **Optional strata:** the same monthly outcome and risk-set grain by sex and predeclared age bands, including 65–69 and 70–74 if governed release permits.
5. **Weather lock:** Hogan-approved HM/CM reference rules before those indicators leave the supplement.
6. **Only then** re-open daily designs if daily or weekly governed outcomes exist; do not retry rich DLNM at $n = 132$ months after M|D failure.
7. **Prediction / ensemble** only after the governed analysis panel is present in the runtime environment and Gate 3 posture is set.

This sequence improves readiness without inventing missing files.

11.1 Minimum viable “next paper” versus “next methods note”

If the team wants a journal manuscript from current files only, the viable thesis is aggregation-aware inference and thermal-definition uncertainty, as in the rewritten manuscript. That paper can be scientifically complete under the recommended Option A posture, or under Option B if the team prefers a cautious qualitative summary.

If the team wants a stronger morbidity claim, the binding constraint is data, not prose. Stroke delivery, risk-set denominators, and dictionary confirmation change the estimand. Writing cannot substitute for them.

If the team wants daily lag claims, the binding constraint is outcome grain. Monthly sums plus failed M|D calibration are not a temporary inconvenience; they are a refusal. Weekly or daily governed outcomes would reopen that door.

11.2 Reproducibility obligations that should travel with any next delivery

Any new governed file should arrive with:

- SHA-256 and row inventory in a data receipt;
- explicit event-construction text;
- provenance label on every analysis row;
- an updated claim ledger rather than orphan coefficients;
- release validation that recomputes multiplicity and checks table-to-ledger identity.

The present CHD/HF release already implements that pattern (`cvd_real_release_validation.md`, 29/29). Future deliveries should not regress to unlabelled CSVs.

12. Conclusion

The delivered science is a governed monthly CHD/HF first-hospitalisation panel under thermal-definition uncertainty and aggregation-aware inference. Continuity estimates show exploratory CHD hot-night and HF cold-day associations with $q = 0.192$, material CHD residual dependence, and SE discordance that forbids selecting NW6 because it excludes the null. Constrained daily-exposure recovery failed calibration and is refused. Daily mortality attributable fractions and modelled excess deaths remain complementary literature baselines, not substitutes for these morbidity count ratios. Gate 3, authorship, dictionary confirmation, stroke delivery, weather lock, and dissemination remain human decisions. The correct next step is to decide those items and, where needed, request the next governed data layer—not to invent a stronger claim from the present release.

Closing inventory for the supervision meeting

Bring to the meeting:

1. This integrated report and the journal manuscript draft.
2. Table 2 and Table 4 from `outputs/release_chd_hf/tables/`.
3. Figure 3 (core forest) and Figure 5 (SE ladder).
4. The F1.2 decision report and the five weather reproduction checks.
5. The open-decision list in Section 10, with Option A recommended and B/C retained as alternatives.

Do not bring synthetic calibration coefficients as if they were CHD/HF effects. Do not bring a frozen headline that the team has not chosen.

The meeting's success criterion is a written decision on Gate 3 posture and on the next governed-data request, not a longer coefficient list.

Internal provenance (not bibliographic references)

`knowledge/2026-08-10_correspondence_scientific_contract.md`; `reports/data_receipt_2026-08-07.md`; `reports/gate2_qc_close_2026-08-07.md`; `analysis_plan/final_reanalysis_protocol_2026-08-10.md`; `analysis_plan/final_claim_tiers.yml`; `outputs/reports/cvd_real_release_validation.md`; `outputs/calibration_md/md_calibration_f1_2_decision_report.md`.

Bibliographic citations are rendered by Pandoc `citeproc` from `literature/references.bib`.

Appendix A — Artifact map (short)

Need	Path
Core tables	outputs/release_chd_hf/tables/table1_*.csv ... table4_*.csv
Claims	.../tables/claim_ledger_v2.csv
Figures 1-5	.../figures/figure{1-5}_*.png
Supplement diagnostics	.../supplement/
Methods feasibility	.../supplement/methods_feasibility/
Journal draft	manuscript/chd_hf_thermal_associations_2013_2023.md
Supplement draft	manuscript/chd_hf_supplement.md
Authorship status	manuscript/authorship_and_governance_decisions.md

See also [README.md](#) in this folder for compile notes.

Appendix B — Deliberately excluded claims

The report excludes five claim classes on purpose: a confirmatory CHD-hot / HF-cold discovery; any real daily M|D coefficient; stroke or principal-diagnosis results; cohort incidence without the risk set; and mortality attributable fractions or modelled deaths presented as morbidity effects. These are scientific refusals, not unfinished prose.

References

- Basagaña, Xavier, and Joan Ballester. 2024. “Unbiased Temperature-Related Mortality Estimates Using Weekly and Monthly Health Data: A New Method for Environmental Epidemiology and Climate Impact Studies.” *The Lancet Planetary Health* 8 (10): e766-77. [https://doi.org/10.1016/S2542-5196\(24\)00212-2](https://doi.org/10.1016/S2542-5196(24)00212-2).
- . 2026. “Unbiased Estimates Using Temporally Aggregated Outcome Data in Time Series Analysis: Generalization to Different Outcomes, Exposures, and Types of Aggregation.” *Epidemiology* 37 (1): 16-20. <https://doi.org/10.1097/EDE.0000000000001923>.
- Chan, Emily Y. Y., William B. Goggins, Janice S. K. Yue, and Poyi Lee. 2013. “Hospital Admissions as a Function of Temperature, Other Weather Phenomena and Pollution Levels in an Urban Setting in China.” *Bulletin of the World Health Organization* 91 (8): 576-84. <https://doi.org/10.2471/BLT.12.113035>.
- Goggins, William B., Emily Y. Y. Chan, and Chun-Yuh Yang. 2013. “Weather, Pollution, and Acute Myocardial Infarction in Hong Kong and Taiwan.” *International Journal of Cardiology* 168 (1): 243-49. <https://doi.org/10.1016/j.ijcard.2012.09.087>.
- Goggins, William B., Jean Woo, Suzanne Ho, Emily Y. Y. Chan, and P. H. Chau. 2012. “Weather, Season, and Daily Stroke Admissions in Hong Kong.” *International Journal of Biometeorology* 56 (5): 865-72. <https://doi.org/10.1007/s00484-011-0491-9>.
- Guo, Yi Tong, Ka Hung Chan, Hong Qiu, Eliza Lai-Yi Wong, and Kin-Fai Ho. 2024. “The Risk of Hospitalization Associated with Hot Nights and Excess Nighttime Heat in a Subtropical Metropolis: A Time-Series Study in Hong Kong, 2000-2019.” *The Lancet Regional Health - Western Pacific* 51: 101168. <https://doi.org/10.1016/j.lanwpc.2024.101168>.
- Guo, Yi Tong, Yingxin Li, Kin-Fai Ho, and Ka Hung Chan. 2025. “Aggravated Risks of Emergency Hospitalizations Associated with Temperature Amid Elevated Ambient Air Pollution: Evidence from a 20-Year Time-Series Study in Hong Kong.” *Environmental Science & Technology* 59 (50): 27107-17. <https://doi.org/10.1021/acs.est.5c10594>.
- Guo, Yuming, Antonio Gasparrini, Ben G. Armstrong, Benjawan Tawatsupa, Aurelio Tobias, Eric Lavigne, Micheline de Sousa Zanotti Stagliorio Coelho, et al. 2017. “Heat Wave and Mortality:

- A Multicountry, Multicommunity Study." *Environmental Health Perspectives* 125 (8): 087006. <https://doi.org/10.1289/EHP1026>.
- Lazarus, Eben, Daniel J. Lewis, James H. Stock, and Mark W. Watson. 2018. "HAR Inference: Recommendations for Practice." *Journal of Business & Economic Statistics* 36 (4): 541-59. <https://doi.org/10.1080/07350015.2018.1506926>.
- Li, Changyu, Wenmin Wei, Pak Wai Chan, and Jianping Huang. 2025. "Heatwaves in Hong Kong and Their Influence on Pollution and Extreme Precipitation." *Atmospheric Research* 315: 107845. <https://doi.org/10.1016/j.atmosres.2024.107845>.
- Liu, Jingwen, Alana Hansen, Blesson Varghese, Zhidong Liu, Michael Tong, Hong Qiu, Linwei Tian, et al. 2020. "Cause-Specific Mortality Attributable to Cold and Hot Ambient Temperatures in Hong Kong: A Time-Series Study, 2006-2016." *Sustainable Cities and Society* 57: 102131. <https://doi.org/10.1016/j.scs.2020.102131>.
- Liu, Zhenyuan, Chao Ren, Jingwen Liu, Yurika Kawasaki, and David Makram Bishai. 2026. "Modelling the Excess Mortality Associated with Heat Waves in Hong Kong: 2014-2023." *medRxiv*. <https://doi.org/10.64898/2026.03.05.26347683>.
- Wang, Dan, Kevin Ka-Lun Lau, Chao Ren, William Bernard Goggins, Yuan Shi, Hung Chak Ho, Tsz-Cheung Lee, Lap-Shun Lee, Jean Woo, and Edward Ng. 2019. "The Impact of Extremely Hot Weather Events on All-Cause Mortality in a Highly Urbanized and Densely Populated Subtropical City: A 10-Year Time-Series Study (2006-2015)." *Science of the Total Environment* 690: 923-31. <https://doi.org/10.1016/j.scitotenv.2019.07.039>.
- Yang, Zihan, Yuchen Wei, Xiaoting Jiang, Conglu Li, Guozhang Lin, Yawen Wang, and Ka Chun Chong. 2025. "Association of Cold Weather and Influenza Infection with Stroke: A 22-Year Time-Series Analysis." *International Journal of Biometeorology* 69 (5): 963-73. <https://doi.org/10.1007/s00484-025-02870-2>.